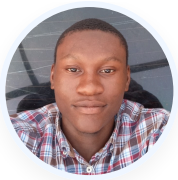
Abdulrahim M Ahmed



Software Developer & Electrical Engineer | AI Engineer

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github.com/Designbrainy | linkedin.com/in/ideahitz | behance.net/abdvrraheem

PROFESSIONAL SUMMARY

Resourceful **Software Developer** and **Electrical Engineering** student at Ahmadu Bello University, combining rigorous engineering principles with modern full-stack development and applied artificial intelligence. Proven track record of designing and deploying functional digital products—from generative AI study platforms and image compression engines to fintech and telecom utilities. Adept at bridging clean user experience with resilient backend architectures, with a dedicated focus on building scalable solutions that solve pressing real-world challenges across African contexts.

TECHNICAL & PROFESSIONAL SKILLS

Programming Languages	Python, JavaScript (ES6+), TypeScript, HTML5, CSS3, SQL
AI & Data Science	Google Gemini API, LLM Prompt Engineering, Natural Language Processing (NLP), NLTK, Sentiment Analysis, pandas, matplotlib, Jupyter
Frontend Engineering	React, TypeScript, Vite, Progressive Web Apps (PWA), Responsive UI/UX Design, CSS3 Animations
Backend & Cloud	Node.js, Express.js, Flask, Netlify Functions (Serverless), RESTful APIs, Drizzle ORM, Docker, Render, Fly.io
Fintech & Compliance	Paystack API, Virtual Accounts, Digital Wallet Float, NIMC & NCC Identity Verification Workflows
Design & Tooling	Brand Identity Design, UI/UX Prototyping, Git, GitHub, VS Code, Postman, Linux/CLI

FEATURED ENGINEERING & AI PROJECTS

EE Scholar AI — *AI-Powered Academic Tutor & Engineering Assistant*

Live Demo GitHub

JavaScript • Netlify Functions • Google Gemini API • Drizzle ORM • PWA • CSS3

- Engineered an interactive AI tutoring application serving Electrical Engineering undergraduates at Ahmadu Bello University, Zaria.
- Integrated **Google Gemini API** with structured system prompting and curriculum retrieval to answer complex technical engineering queries and evaluate study files.
- Implemented multi-modal study tools including voice-based tutoring interactions and an automated Nigerian 5-point GPA calculator.
- Packaged as an installable **Progressive Web App (PWA)** with offline support and serverless Netlify microservices for low-latency delivery.

MTN Village Portal — *Telecom Vending & Identity Registration Platform*

Live Demo GitHub

Node.js • Express.js • JavaScript • Paystack API • Render

- Developed a telecom automation and government citizen identity booking portal designed for the Birnin Gwari community.
- Integrated the **Paystack API** to enable instant multi-network data/airtime vending, automated virtual accounts, and digital wallet float balances.
- Engineered structured booking pipelines compliant with **NIMC** and **NCC** regulatory frameworks for NIN and BVN enrollments.
- Designed cryptographic-style verifiable digital receipts to prevent transaction forgery across local retail kiosks.

Distill — *Privacy-First Image Optimization Microservice* Live Demo GitHub

Python • Flask • Pillow (PIL) • Docker • Fly.io • Render

- Engineered an image compression service capable of processing JPEG, PNG, and WebP assets up to 50MB with strict quality preservation.
- Implemented target byte-size threshold calculations while ensuring PNG alpha transparency channel integrity.
- Built zero-tracking, temporary in-memory buffering mechanisms ensuring user data is never stored or tracked.
- Containerized application dependencies with **Docker** for reproducible, production-ready cloud deployment.

NaijaScholar AI — *Intelligent Academic Assistant for Nigerian Curricula* GitHub

React • TypeScript • Vite • Google Gemini API • Netlify

- Constructed a type-safe single-page application integrating Google Gemini to generate curated academic mentoring and scholarship guidance.
- Engineered performant frontend components using React and Vite, optimizing mobile web responsiveness and data payload size.

Twitter Sentiment Analysis Pipeline — *NLP & Social Opinion Mining* GitHub

Python • NLTK • pandas • matplotlib • Jupyter Notebook

- Implemented a complete natural language processing workflow processing over 10,000 tweets with tokenization, stop-word removal, and word stemming.
- Generated empirical statistical summaries and graphical sentiment distributions using pandas and matplotlib.

DESIGN & CREATIVE EXPERIENCE

Brand & Visual Identity Designer — *Freelance & Product Design* Behance Portfolio

- Developed visual identity guidelines, typography systems, and high-fidelity interface layouts for technology projects and brand clients.
- Unified engineering logic with accessible human-centered UX design to maximize adoption across web applications.

EDUCATION

Bachelor of Engineering (B.Eng.) in Electrical Engineering — Ahmadu Bello University (ABU), Zaria, Nigeria

Relevant Coursework: Software Systems, Data Structures & Algorithms, Digital Electronics, Circuit Analysis, Signals & Systems, Control Engineering, Applied Mathematics.
Academic Focus: Practical AI/ML integration, embedded systems, and scalable software solutions.

ADDITIONAL DETAILS & HIGHLIGHTS

Languages	English (Fluent / Professional Proficiency), Hausa (Native)
Core Focus	Applied AI & Generative Systems, Privacy-First Architecture, Local Telecom Infrastructure, Community Impact
Engineering Approach	First-principles problem breakdown, empirical testing, modular architecture, and human-centered design.